



RAINFALL PATTERN ANALYSIS USING CONTINUOUS PROBABILITY DISTRIBUTIONS



Exploratory analysis of monthly and annual rainfall across Indian districts

Guided by:
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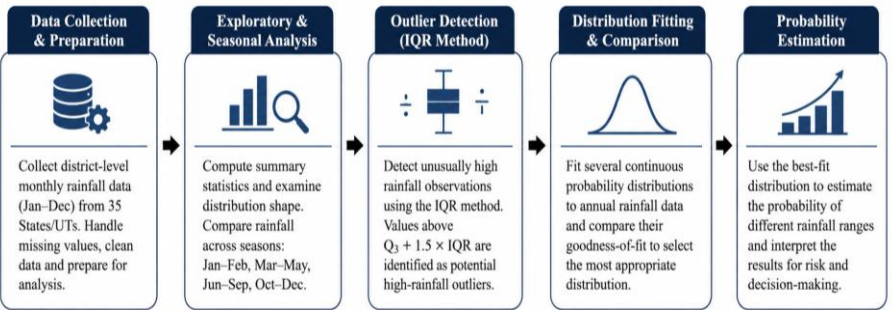
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AIM & OBJECTIVES

To analyze rainfall patterns across Indian districts and estimate rainfall probabilities using continuous distributions.

- Analyze rainfall patterns across Indian districts and seasons.
- Summarize rainfall using statistical measures and distribution analysis.
- Identify unusual rainfall values using the IQR method.
- Use continuous probability distributions to estimate rainfall probabilities

METHODOLOGY



KEY FINDINGS

- Annual rainfall is strongly right-skewed.
- June–September is the dominant rainfall period.
- IQR highlights unusually high annual-rainfall observations.
- High variability exists across districts.

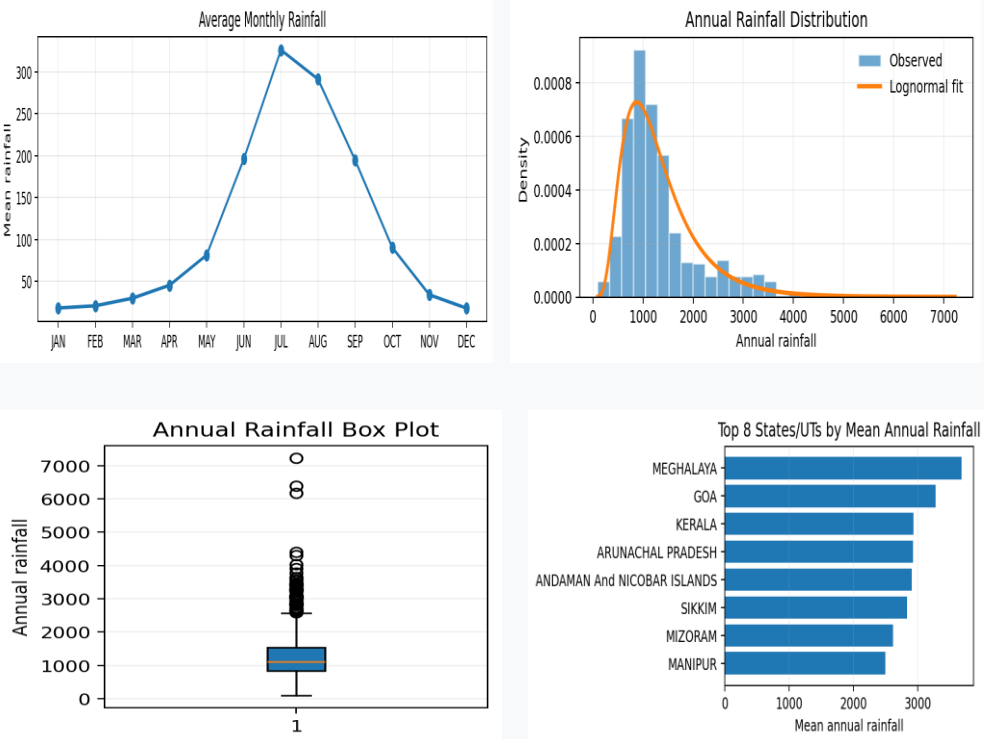
DATASET

Indian District Rainfall Dataset

641 district records | 35 States/UTs | Rainfall in mm
Variables include State/UT, District, Jan–Dec rainfall, seasonal totals and annual rainfall.

Source: Kaggle dataset

RESULTS



STATISTICAL INTERPRETATION

Continuous probability view

- Rainfall distribution is not symmetric.
- Skewness = 2.19, indicating strong positive/right skewness
- A small number of high-rainfall districts create the long upper tail.
- The fitted distribution helps convert rainfall observations into probability-based estimates.
- These estimates can support rainfall-risk and resource-planning decisions.

CONCLUSION

Indian district rainfall is highly variable, seasonal and strongly right-skewed. The combination of descriptive statistics, IQR analysis and probability distributions provides a useful framework for understanding rainfall variability and estimating rainfall likelihoods.

FUTURE SCOPE / ENHANCEMENTS

- Add multi-year records to study temporal change.
- Compare districts using regional probability models.
- Evaluate more distributions and goodness-of-fit tests.
- Develop drought and extreme-rainfall alerts.
- Integrate climate and geographic variables.

1347.0

Mean annual rainfall

1116.2

Median annual rainfall

2.19

Skewness